

Bogdan Løv-Hansen, PhD

Researcher at the [NTNU UAV Icing Lab](#) | Engineer at [UBIQ Aerospace](#)

PhD in Engineering Cybernetics (NTNU, 2026), specializing in UAV icing: aerodynamic modeling, system identification, and rigid-body simulation of fixed-wing UAVs, and the model-based detection algorithms built on them. Broader background in guidance, navigation, and control (GNC) for autonomous systems, including state estimation and statistical fault detection. Experience applying deep neural networks for image classification, and development of a model-based "virtual" detection sensor for UAV icing during the PhD. Appointed member of the NATO STO AVT-388 panel and selected as a 2024 Arctic Frontiers Emerging Leader.



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PUBLICATIONS

First author unless noted.

- 2026 System Identification of Leading-Edge Rime-Ice Effects on a Small Fixed-Wing Drone. *The Aeronautical Journal* (accepted)
- 2025 [Modeling and Identification of a Small Fixed-Wing UAV Using Estimated Aerodynamic Angles](#). *CEAS Aeronautical Journal*
- 2024 [Time-Domain System Identification and Validation of Small Fixed-Wing UAV Dynamics](#). *AIAA Aviation Forum*
- 2023 [UAV Icing: A Survey of Recent Developments in Ice Detection Methods](#). *IFAC World Congress*
- 2023 [Identification of an Electric UAV Propulsion System in Icing Conditions](#). *SAE Technical Paper Series*
- 2023 [UAV Icing: Development of an Ice Protection System for the Propeller of a Small UAV](#). *Cold Regions Science and Technology* (co-author, Müller et al.)
- 2022 [UAV Icing: Ice Shedding Detection Methods for an Electrothermal De-Icing System](#). *AIAA Aviation Forum*

WORK EXPERIENCE

NTNU, Trondheim Dept. of Engineering Cybernetics MAR 2026 - PRESENT	Researcher Developing ice protection solutions for green aviation under the Zero Ice Shield research project.
AUG 2021 - MAR 2026	Ph.D. Candidate Researched and developed autonomous detection algorithms for in-flight icing on UAVs.
AUG 2022 - JAN 2026	Scientific Assistant TTK4190 Guidance, Navigation and Control of Vehicles . Managed course assignments, supervised students, and graded exams.
AUG 2020 - APR 2021	Student Assistant TTK4260 Multivariate Analysis and Machine Learning Methods and TTK4190.
UBIQ Aerospace, Trondheim MAR 2023 - PRESENT	Modeling and Simulation Engineer Delivering experimental and simulation-based analyses to support engineering operations in UAV icing conditions.
DTU Electro, Denmark JAN 2025 - JUL 2025	Visiting Researcher, CREA group Conducted statistical model-based real-time detection of propeller icing on UAVs.
Aker BioMarine, Oslo JUN 2020 - AUG 2020	Summer Intern Implemented a convolutional neural network (CNN) using PyTorch for plankton image classification. Trained models remotely on GCP, outperforming baseline manual feature engineering methods. results shared with the Institute of Marine Research.
Ingeniarius, Portugal JUL 2019 - SEP 2019	RobotCraft 2019, Internship/Summer-Course Designed software for a wheeled robot to autonomously map and solve a maze-navigation task. Implemented the navigation algorithm fusing 2D Lidar scans and motor encoders using ROS and the A* algorithm.

PROFESSIONAL AFFILIATIONS & AWARDS

NATO STO MAR 2026 - PRESENT	Member of the Applied Vehicle Technology (AVT-388) Group Operation of Unmanned Aerial Vehicles (UAVs) in Icing Environments.
Emerging Leader 2024	Issued by Arctic Frontiers . Selected for a 1-week mentorship program for young global leaders addressing Arctic challenges.

EDUCATION

NTNU, Trondheim AUG 2015 - JUL 2021	M.Sc. in Cybernetics and Robotics Master's thesis: Electrothermal Ice-Shedding Detection for UAVs. Specialization Project: Lidar-based Positioning for Docking of Autonomous Ships.
KTH, Sweden AUG 2018 - JUN 2019	Exchange Student in Stockholm Coursework focus: Signal processing, non-linear control, MIMO-control, and embedded control systems.

STUDENT ORGANISATIONS

NTNU Ascend SEP 2019 - AUG 2020	Aerial Robotics, Team 2020 Implemented lidar-based obstacle avoidance and path-following control.
KTH FS SEP 2018 - JUN 2019	Formula Student, Team 2019 Worked with vehicle component tuning and sensor data analysis.

COMPUTER SKILLS

LANGUAGES:	MATLAB, Python, C++, Bash/Shell.
FRAMEWORKS & TOOLS:	Simulink, PyTorch, ROS, Docker, Git (GitLab/GitHub), Linux (Ubuntu), Google Cloud Platform (GCP).
TECHNIQUES:	Modeling and system identification, autonomous vehicle guidance navigation and control (GNC), state estimation/Kalman filtering, data analysis and visualization, convolutional neural networks (CNN).

OTHER WORK EXPERIENCE

Forsvaret SEP 2014 - AUG 2015	Military service in Norwegian Navy (Haakonsværn) Served aboard KNM Rauma and KNM Hitra.
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LANGUAGES & CERTIFICATES

NORWEGIAN:	Fluent	2022	Driving license, class - A and B
ENGLISH:	Professional Proficiency	2012	Forklift license, class - T4
UKRAINIAN:	Conversant		